Course Title: Computer Organization and Programming

Course Code: MC112

Course Credit: 3-0-0-3

Program: B.Tech. (MnC)

Prerequisite: None

Category: Core

Course Content: This course provides an introduction to computer organization, assembly language programming, and high level language programming using C. Topics include memory unit, arithmetic and logic unit, control unit, addressing modes and instructions for assembly language programming, along with C programming which covers data types, conditions and loops, arrays, functions, structures, union and pointers.

Computer organization: component of computer, memory unit, computation unit, control unit, input/output unit, Bits, data types, operation, logic structures, Von Neumann model, language

Programming: Basic Syntax, Memory Segments, Registers, System Calls, Addressing Modes, Variables, Constants, Arithmetic and logical Instructions, Conditions and Loops, Numbers, strings and arrays

C Programming: Brief Overview, Basic Syntax, Program Structure, Data Types, Variables, Constant, Conditions and Loops, Arrays, Function, Pointers, Structures and Unions.

Text Book:

1. Yale Patt and Sanjay Patel, Introduction to Computing Systems: From Bits & Gates to C & Beyond, 2nd Edition, McGraw Hill, 2003.

Reference Books:

- William C. Gear, Computer Organization and Programming, McGraw Hill, 2016.
- William Stallings, Computer Organization and Architecture, 10th edition, Pearson, 2016.
- Carl Hamacher, Zvonko Vranesic and Safwat Zaky, Computer Organization, 5th Edition, McGraw Hill, 2011.
- James L. Peterson, Computer Organization and Assembly Language Programming, Academic Press, 1978.
- Paul Deitel and Harvey Deitel, C How to Program, Pearson; 8th edition, 2015.
- Brian W. Kernighan and Dennis Ritchie, The C Programming Language, Pearson Education, 2nd edition, 2015.
- E. Balagurusamy, Programming in ANSI C, McGraw Hill, 7th edition, 2017.

Evaluation Components:
Class Test – 15%
Mid semester evaluation – 35%
End semester evaluation – 50%

Course Outcomes:

At the end of the course student will be able to:

- Design instruction set architecture
- Apply the concepts of digital logic to design the basic building blocks of a computer
- Understand the internal architecture of the CPU, memory hierarchy, assembly language instructions and addressing modes and timing analysis of instructions
- Write efficient C programs and analyze them to improve the space and time complexity

P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**	**P10**	**P11**	**P12**
X	X	X		X	X						X